Name _____ _____ Baggett/Fall2026/207/Quiz5

first name last name

Score

No Calculators. / 20

Questions 1 and 2. Circle **True** or **False**. Then, in **one sentence**, say why — if the statement is false, the fastest way is usually to give an example that breaks it. **1 point for the circle, 1 for the reason.**

1. (**2** pts) If f and g are differentiable, then $\frac{d}{dx}[f(g(x))] = f'(g(x))g'(x)$.

True **False**

Why? _____ /2

2. (**2** pts) To differentiate implicitly you must first solve the equation for y.

True **False**

Why? _____ /2

3. (**4** pts) **Derive** $\frac{d}{dx}[\csc x]$ starting from $\csc x = \frac{1}{\sin x}$. Use the Quotient Rule and **show every step**. Quoting the formula earns nothing.

_____ /4

4. (**2** pts) Differentiate $f(x) = \sqrt{x^2 + 9}$. You do not need to simplify.

_____ /2

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5. (**3** pts) Differentiate $g(t) = e^{-2t}\cos(3t)$. You do not need to simplify.

/3

6. (**4** pts) Find $\dfrac{dy}{dx}$ by **implicit differentiation**:

$$x^2 + xy + y^3 = 5$$

/4

7. (**3** pts) Differentiate $y = x^2 \sec x$.

/3

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